INVENT FOR THE PLANET 2026

Invent for the Planet (IFTP) is a global educational initiative that brings together engineering students from universities around the world to address real, pressing challenges facing society. Through an intensive 48-hour experience, the program promotes **problem-driven innovation**, intercultural collaboration, and the practical application of engineering knowledge.

Unlike traditional hackathons, Invent for the Planet places its primary emphasis on **educational outcomes**, fostering the development of core competencies such as systems thinking, stakeholder empathy, applied research, multidisciplinary teamwork, and purposeful prototyping.

The program is strongly aligned with the **United Nations Sustainable Development Goals (SDGs)**, encouraging solutions that balance technical feasibility, social impact, environmental responsibility, and economic context.

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1. Central Theme – Invent for the Planet 2026

Reimagining Food Systems for a Resilient Planet

Theme Context and Rationale

Global food systems are under unprecedented pressure. Population growth, climate change, water scarcity, soil degradation, disruptions in global supply chains, and social and economic inequalities increasingly challenge the world's ability to produce, distribute, and consume food in a secure, efficient, and sustainable manner.

At the same time, food systems represent one of the greatest opportunities for high-impact innovation in the 21st century. They sit at the intersection of engineering, technology, data science, energy, water, logistics, public policy, and human behavior, providing fertile ground for solutions that transcend geographical, cultural, and disciplinary boundaries.

The 2026 edition of Invent for the Planet invites students to **fundamentally rethink food systems**, moving beyond traditional agricultural production to consider the entire value chain—from natural resource use to distribution, from producers to consumers, and from waste to circularity. The focus is not merely on producing more food, but on **building**

resilient, inclusive, and sustainable food systems capable of adapting to climatic, economic, and social shocks.

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2. Challenge Structure (Need Statements)

The challenges for Invent for the Planet 2026 are presented in the form of **broad, intentionally open-ended Need Statements**.

During the 48-hour competition, teams are expected to:

- Conduct **primary research** (interviews, observation, stakeholder validation);
- Perform **secondary research** (public data, literature, benchmarks, case studies);
- Clearly define **the specific problem they choose to address**;
- Develop and test a **functional or conceptual prototype** that responds to the identified problem.
- Present a **pitch to a panel of judges** who will evaluate the innovation and future potential of the developed product.

The Need Statements **do not prescribe solutions**. Instead, they frame **complex problem contexts**, encouraging discovery, interpretation, and evidence-based decision-making.

Official Need Statements – Invent for the Planet 2026

Need Statement 1 – Resilience in Food Production

Food producers across diverse climatic and socioeconomic regions are increasingly exposed to extreme weather events, climate variability, soil degradation, and resource constraints, which threaten the stability, reliability, and long-term sustainability of food production systems.

How might food production systems be designed or adapted to improve resilience and reliability under increasing climate and resource uncertainty, while remaining economically viable and environmentally responsible?

Need Statement 2 – Food Loss, Waste, and System Inefficiency

A substantial share of food produced globally is lost or wasted at different stages of the food system—including harvesting, storage, transportation, processing, retail, and consumption—due to infrastructural limitations, information gaps, and misaligned incentives among stakeholders.

How might food systems be restructured or supported to significantly reduce food loss and waste while improving efficiency, accessibility, and sustainability across the value chain?

Need Statement 3 – Access, Equity, and Smallholder Inclusion

Small-scale farmers and marginalized communities in both developing and industrialized food systems often face limited access to technology, data, markets, financial services, and infrastructure, restricting their ability to participate competitively and equitably in modern food systems and to adapt to environmental and economic change.

How might food systems be designed to increase access, inclusion, and economic resilience for smallholder farmers and vulnerable communities without increasing complexity or dependency?



Need Statement 4 – Water, Energy, and Food Nexus

Food production and distribution systems increasingly compete with urban, industrial, and environmental demands for limited water and energy resources, particularly in regions facing chronic scarcity, rising costs, and climate stress.

How might food systems better integrate water and energy considerations to ensure sustainable, efficient, and resilient food production under conditions of resource scarcity?

Need Statement 5 – Transparency and Trust in Food Systems

As food supply chains become more global, complex, and fragmented, consumers, producers, and regulators face growing challenges in ensuring transparency, traceability, accountability, and trust related to food origin, quality, safety, and environmental impact.

How might food systems be designed to increase transparency and trust among stakeholders while balancing cost, scalability, and data reliability?

Need Statement 6 – Household-Centered Circular Material Systems

Households generate a wide range of materials—including packaging, textiles, electronics, and organic waste—that often lose their economic, environmental, and social value at the point of disposal. Despite the existence of recycling and waste management systems, fragmented infrastructure, limited incentives, information gaps, and misaligned responsibilities among consumers, collectors, producers, and regulators result in low material recovery rates and continued reliance on landfilling and incineration.

How might household-centered material systems be designed to recognize, retain, and fairly distribute material value while enabling transparent, inclusive, and scalable circular flows across communities and the wider materials economy?

3. Alignment with the United Nations Sustainable Development Goals

The challenges of Invent for the Planet 2026 are directly aligned with the following SDGs:

- **SDG 2** – Zero Hunger and Sustainable Agriculture
- **SDG 6** – Clean Water and Sanitation
- **SDG 12** – Responsible Consumption and Production
- **SDG 13** – Climate Action
- **SDG 15** – Life on Land
- **SDGs 8 and 9** – Decent Work, Economic Growth, Innovation, and Infrastructure

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4. Expected Educational Outcomes

By the conclusion of Invent for the Planet 2026, participating students are expected to have developed:

- The ability to **define and frame complex, real-world problems**;
- Competence in **applied research and user validation**;
- Strong **systems-thinking** and trade-off analysis skills;
- **Rapid prototyping** capabilities;
- Technical communication and impact-driven storytelling skills;
- Ethical, social, and environmental awareness in the practice of engineering.

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5. Final Remarks

Invent for the Planet 2026 reaffirms the program's commitment to educating engineers who are prepared to operate in an interconnected, uncertain, and rapidly changing world. By reimagining global food systems, participants will not only develop technical solutions, but also deepen their understanding of the role of engineering in building a more resilient, equitable, and sustainable future.